

Kuan-Chia, Chen

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Education

Oregon State University <i>Master in Computer Science</i>	Sep 2023 – Jun 2026 Corvallis, OR
National Taipei University of Technology <i>B.S. in Electrical Engineering</i>	Sep 2019 – Jun 2023 Taipei, Taiwan

Research Experience

Dynamic Robotics and Artificial Intelligence Laboratory <i>Graduate Research Assistant-Prof. Alan Fern</i>	Corvallis, OR Dec 2024 – Jun 2026
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Publications

Humanoid Hanoi: Investigating Shared Whole-Body Control for Skill-Based Box Rearrangement <i>Under review for RSS 2026</i>	2026
Minku Kim [†] , Kuan-Chia Chen [†] , Aayam Shrestha, Li Fuxin, Stefan Lee and Alan Fern	

Projects

Point Cloud to Action <i>Robotics Project (PPO + LSTM, Isaac Lab)</i>	Corvallis, OR 2025
Built a hierarchical robotic manipulation system using 3D point cloud observations for robust box pick up across diverse positions, orientations, and sizes. Trained a PPO-based policy with a two-layer LSTM in Isaac Lab on the Digit V3 humanoid robot, allowing the model to generate continuous control for sequential manipulation tasks.	
Diffusion Policy for Robot Control <i>Robotics Project (Diffusion, Mujoco)</i>	Corvallis, OR 2025
Developed a diffusion U-Net policy for robotic manipulation, generating control actions conditioned on object properties (position, orientation, size) and robot state. Trained on 100 VR-collected human demonstration trajectories for one box and validated in MuJoCo, achieving robust box pick up from limited data.	
Keypoint Mimic Policy for VR Control <i>Robotics Project (VR, Teleoperation, PPO + LSTM, Mujoco)</i>	Corvallis, OR 2025
Developed an end-to-end keypoint mimic framework for robot teleoperation using a VR headset, mapping human hand keypoints and locomotion inputs to robot control commands.	

Skills

Robotics & Simulation: MuJoCo, Isaac Lab
Machine Learning: Reinforcement Learning (PPO, Diffusion Policy)
Programming: Python, C++
Embedded & Control Systems: Arduino, PLC (Mitsubishi, Omron), Siemens LOGO!
Building Automation: KNX Smart Home Systems

Awards

First Place - Taiwan National Senior Secondary School Industrial Control Competition	2018
Silver Medal - Taiwan National Electrical Installations Competition (Central Region)	2018